



Feature Specification & User Flows

SUPER ADMINS · COMPANY ADMINS · KIOSK OPERATORS · EMPLOYEES · SYSTEM ENGINES

Core features, how each works, and the step-by-step flow for every user type involved.

01 — STATION KIOSK CHECK-IN & BIOMETRIC FACE VERIFICATION

Workers clock in or out directly at a dedicated station kiosk terminal using contactless facial biometrics with instant anti-spoof verification.

EMPLOYEE

1. Step in front of the Station Kiosk camera frame
2. In Privacy Mode: Type the first few characters of your name or email into the lookup bar (or allow instant camera detection)
3. Align face within the on-screen targeting reticle
4. System executes an automatic anti-spoof liveness check (head tilt / eye blink / texture validation)
5. Receive immediate visual confirmation: Green checkmark, employee name, avatar, and timestamp
6. Terminal displays shift status and plays an audible chime ("Approved - Clocked In")
7. Interface automatically resets after 7 seconds for the next worker in queue

KIOSK OPERATOR / SUPERVISOR

1. Authenticate and unlock the Station Kiosk using Station Credentials or NFC Master Key
2. Monitor the check-in queue during peak morning rush or shift changeover
3. Initiate Assisted Check-In to allow manual PIN or password verification if worker has camera issues
4. Review flagged low-confidence scans and authorize supervisor overrides when necessary

COMPANY ADMIN / HR MANAGER

1. Navigate to Employee Management and initiate Biometric Enrollment
2. Capture 3 reference facial angles to generate a 512-dimensional embedding vector
3. Configure Station Matching Thresholds (0.60 strict cosine similarity or 0.70 standard)

02 — ZERO-DOWNTIME OFFLINE ATTENDANCE & BACKGROUND OUTBOX SYNC

The Station Kiosk operates completely offline during network blackouts, queuing records locally and synchronizing seamlessly upon reconnection.

EMPLOYEE

1. Step up to the Station Kiosk during an internet connectivity outage
2. Look at the camera or enter your standard PIN
3. The kiosk matches credentials against the locally encrypted on-device employee cache
4. Receive immediate visual confirmation: "Approved (Offline Queued)", along with shift status
5. On-screen active headcount increments immediately (e.g. 6/7 updates to 7/7 present)
6. Step away with complete assurance that proof of work is locked to the exact second

KIOSK OPERATOR / SUPERVISOR

1. Observe status bar displaying the yellow "Offline Mode — Running Locally" badge
2. Inspect the Offline Outbox counter showing queued punches (e.g. "14 records pending")
3. Continue shift operations completely uninterrupted; no worker is turned away
4. Upon reconnection, watch status turn green ("Online") and outbox automatically flush

SYSTEM / BACKGROUND SYNC ENGINE

1. Detect network failure and route punches to IndexedDB attendance_outbox
2. Maintain 15-second reconnection poll loop with exponential backoff
3. Send idempotent batch payload to POST /api/station/attendance/batch-sync
4. Commit records to PostgreSQL, clear outbox, and broadcast WebSocket alerts

03 — ROTATING 30-SECOND DYNAMIC QR CODE ATTENDANCE

A dynamic cryptographic QR code regenerates every 30 seconds on the kiosk display, allowing anti-proxy mobile scans within validated geofence perimeters.

EMPLOYEE (MOBILE APP)

1. Open the TimeLogic Mobile App upon arriving at the workplace
2. Tap "Scan Station QR" on the mobile dashboard
3. Point phone camera at the rotating QR displayed on the kiosk screen
4. App captures the token, attaches device GPS coordinates & Wi-Fi BSSID, and transmits payload
5. Receive instant haptic confirmation: "Clock-In Verified — Shift Started"

KIOSK DISPLAY TERMINAL

1. Display rotating QR mode with an animated 30-second circular countdown timer
2. Regenerate a new cryptographic TOTP hash every 30 seconds
3. Flash discreet green confirmation banner without interrupting rotation

COMPANY ADMIN / HR MANAGER

1. Enable "Dynamic QR Mode" for specific branches, job sites, or project zones
2. Define authorized Geofence Radius (e.g. 50 meters) and Wi-Fi router BSSIDs
3. View mobile punch logs flagged with precision coordinates and map pins

SYSTEM / CRYPTOGRAPHIC ENGINE

1. Generate short-lived HMAC-SHA256 payload with 30s TTL
2. Validate nonce has not been consumed (prevents replay / screenshot attacks)
3. Verify device coordinates against branch geofence boundary via Haversine formula
4. Commit attendance record and invalidate token immediately

04 — BREAK ROOM & SHIFT LIFECYCLE MANAGEMENT

Workers transition seamlessly between active shift hours, paid meal breaks, and unpaid pauses with automated net billable hours tracking.

EMPLOYEE

1. Step up to the Station Kiosk or open the Mobile App during shift hours
2. Authenticate via Face, PIN, or QR
3. Select "Start Break" (Meal / Rest / Personal)
4. Shift timer pauses billable hours; kiosk displays a blue "On Break" badge
5. Upon returning, scan again and tap "Resume Work"
6. At shift conclusion, tap "Clock Out" to receive total shift & break summary

KIOSK OPERATOR / SUPERVISOR

1. Monitor live floor status: distinct counts for "Active On Shift", "On Break", and "Out"
2. Identify workers exceeding scheduled break allowances via yellow visual flags
3. Assist workers who forgot to resume break through the Supervisor Console

ADMIN / AUTOMATION ENGINE

1. Admin configures Paid vs Unpaid break thresholds (e.g. 30min unpaid lunch)
2. Engine tracks exact timestamps and computes: Net Hours = Shift Duration - Unpaid Breaks
3. Nightly cron closes abandoned open shifts, tagging them AUTO_CLOCKED_OUT

05 — AUTOMATED PENALTY, LATE TIERS & OVERTIME ENGINE

Late arrivals are categorized into customizable grace periods and penalty tiers, calculating automatic salary deductions with transparent waiver workflows.

EMPLOYEE

1. Clock in past the scheduled shift start time
2. Kiosk displays a polite notice: "Clocked in at 08:18 (18 mins late — Tier 1 Late Arrival)"
3. View breakdown of accumulated late minutes and salary impact on personal portal
4. If delay was caused by an approved errand or emergency, tap "Request Late Waiver"
5. Receive real-time push notification once HR approves or denies the waiver

PAYROLL OFFICER / FINANCE

1. Generate monthly attendance-payroll ledger at the end of the pay cycle
2. Review verified overtime, tardiness deductions, and approved waivers
3. Export directly to CSV or sync via API to integrated payroll software

COMPANY ADMIN / HR MANAGER

1. Configure Shift Grace Periods (e.g. 5-minute standard grace before penalty)
2. Set Late Penalty Tiers (Tier 1: warning; Tier 2: 30min deduction; Tier 3: 1hr deduction)
3. Open Disputes & Waivers queue to review pending employee waiver submissions
4. Approve with note (waives deduction immediately) or reject with explanation

SYSTEM / RULES ENGINE

1. Compare punch timestamp against scheduled shift start
2. Calculate late minutes and apply configured company penalty matrix
3. If waiver is approved, update ledger dynamically with immutable audit trail

06 — HARDWARE DEVICE BINDING & FLEET TERMINAL MANAGEMENT

Hardware kiosks and mobile stations are enrolled via cryptographic handshakes, hardware fingerprinting, and remote seat allocations.

KIOSK OPERATOR / IT INSTALLER

1. Power on the hardware tablet or kiosk machine and launch TimeLogic Station App
2. Select "Register New Terminal" and enter the one-time Station Setup Key
3. Name the terminal (e.g. "Factory Gate North — Tablet 02") and grant camera permissions
4. App binds the physical hardware to the tenant and locks into Kiosk Mode

SUPER ADMIN (PLATFORM LEVEL)

1. Monitor global terminal fleet metrics across all client organizations
2. Deploy over-the-air client updates and firmware compatibility patches
3. Track station connection health and API latency across geographic regions

COMPANY ADMIN / IT MANAGER

1. Open "Terminal & Fleet Management" in the Admin Dashboard
2. View all active registered terminals, branch assignments, and heartbeat timestamps
3. Allocate available Station Terminal Seats based on subscription quota
4. If a device is lost or stolen: Tap "Revoke Device" to immediately sever tokens

SYSTEM / AUTHENTICATION SERVICE

1. Extract WebCrypto device key, platform UUID, and hardware fingerprint
2. Issue 30-day Station JWT and 90-day Refresh Token with auto-rotation
3. On revocation: Blacklist station token in Redis/memory cache instantaneously

07 — EXECUTIVE AUDIT LOGS, LIVE WEBSOCKET FEEDS & FRAUD ALERTS

Real-time attendance wallboards provide instant floor visibility, backed by an immutable audit ledger flagging duplicate punches and spoofing.

COMPANY ADMIN / HR EXECUTIVE

1. Open the "Executive Live Feed" on office TV monitors or desktop browser
2. Watch the real-time stream of clock-ins across all corporate branches with zero delay
3. Filter by branch, department, or exception status (late arrivals, overrides)
4. Receive instant desktop alerts if high-severity anomalies occur
5. Generate signed PDF / CSV Statutory Audit Reports for compliance reviews

SUPER ADMIN (PLATFORM OWNER)

1. Access the global platform security and threat intelligence dashboard
2. Track facial anti-spoof rejection rates across all client camera hardware
3. Investigate suspicious API spikes or unauthorized token replay attempts

SYSTEM / SECURITY ENGINE

1. Broadcast attendance event via WebSocket in <50ms to subscribed channels
2. Execute Duplicate Punch Guard (block repeat punches within 60s)
3. Flag Impossible Velocity if worker punches at distant branches too quickly
4. Append administrative actions to tamper-evident audit ledger

08 — MULTI-BRANCH SYNCHRONIZATION & GEOFENCE PERIMETERS

Enterprises manage distributed physical branch locations with localized geofences, regional shift calendars, and cross-branch floating worker access.

FLOATING EMPLOYEE

1. Arrive at a secondary corporate branch location or satellite warehouse
2. Step up to the local station kiosk or open mobile app
3. System recognizes multi-branch floating permissions and verifies face or PIN
4. Shift is recorded against the local branch cost center while syncing to home department

BRANCH SUPERVISOR

1. View branch-specific attendance wallboard displaying local floor headcount
2. Identify visiting floating workers distinguished by a "Visiting Staff" indicator
3. Sign off on local shift handoffs and overtime authorization

COMPANY ADMIN / HR DIRECTOR

1. Define geographical branch perimeters using interactive polygon mapping tools
2. Assign branch-specific work hours, localized holidays, and grace period policies
3. Designate which workers are restricted to home sites vs granted roaming access

SYSTEM / GEOSPATIAL ENGINE

1. Validate GPS fix accuracy before comparing with branch polygon geofence
2. Route attendance records to appropriate local branch and corporate databases
3. Isolate branch sub-admin permissions so managers only see local workforce logs

09 — ENTERPRISE B2B & PAYROLL ERP INTEGRATIONS

Pre-built connectors synchronize verified hours, overtime, and penalty deductions directly into enterprise payroll, HRIS, and ERP platforms.

PAYROLL ACCOUNTANT

1. Open the Payroll Integration Studio at the conclusion of the pay cycle
2. Trigger automated pre-payroll validation check to highlight unresolved waivers
3. Click "Synchronize to Payroll" (Gusto, QuickBooks, Paystack, ADP, SAP)
4. Download cryptographically signed audit certificate for corporate compliance

ENTERPRISE PARTNER ADMIN

1. Configure secure OAuth2 or API token credentials for corporate HRIS connection
2. Map TimeLogic cost centers and shift codes to enterprise general ledger accounts
3. Define automated weekly or monthly synchronization schedules

SYSTEM / INTEGRATION ENGINE

1. Ingest verified shift durations, verified overtime multipliers, and late deductions
2. Transform ledger into target schema (JSON / NACHA / CSV / REST API)
3. Execute transactional sync with automatic retry queues and failure alerts
4. Lock synchronized pay period to prevent retroactive tampering or modifications

TIMELOGIC SOLUTIONS OPERATIONS

1. Maintain high-availability webhook pipelines and ERP connectors
2. Audit integration sync health under enterprise 99.99% uptime SLA

PROPOSED FEATURE

Proposed Feature: AI-Driven Predictive Rostering & Fatigue Detection

UNDER EXPLORATION — NOT YET SCOPED FOR BUILD

A natural extension of TimeLogic's biometric telemetry and shift tracking engine. The system analyzes historical attendance patterns, peak rush hours, absenteeism trends, and overtime accumulation to automatically generate optimized, fatigue-safe shift schedules while proactively warning managers of burnout and labor law violations.

WHY THIS FITS TIMELOGIC

TimeLogic already captures the exact operational truth — when workers arrive, how long they work, how many breaks they take, and when operations experience labor shortages. Predictive Rostering turns historical attendance data into proactive, automated workforce scheduling inside the exact same platform.

EMPLOYEE

1. View AI-recommended shift rosters two weeks in advance in mobile app
2. Receive proactive fatigue alerts (e.g. 48 hours completed across 5 shifts)
3. Request shift swaps; AI validates matching qualifications and overtime safety
4. Confirm or request adjustments to scheduled shifts with one tap

HR PLANNER / OPERATIONS

1. Select scheduling targets: coverage, skill certifications, budget caps
2. Click "Generate Optimized Schedule" to build draft roster in seconds
3. Review proactive burnout flags and overtime accumulation warnings
4. Publish finalized roster with one click to all employee mobile apps

OPEN QUESTIONS TO RESOLVE BEFORE BUILDING

- **Labor Law Variance:** Overtime ceilings and mandatory rest periods vary drastically across operating jurisdictions (Nigerian Labor Act, US FLSA, EU Working Time Directive) requiring modular rule engines.
- **Privacy by Design:** Fatigue estimation must be framed constructively as employee wellness safeguards and supervisor workload limits, never punitive surveillance.
- **Algorithmic Fairness:** Automated scheduling models must be audited to ensure unbiased shift distribution, preventing preferential allocation of premium overtime hours.
- **Offline Roster Caching:** Roster schedules must be pre-cached on Station Kiosks so shift assignments remain fully accessible during offline network disconnects.